01 / CAN A FEED FOLLOW A NEW INTEREST?

Your tastes can change. Can your feed?

We tested how one history-based recommender responds when a profile explores an infrequently liked movie genre. The histories are real; the new likes are simulated.

TARGET REACHED WITHIN FIVE NEW LIKES

1.5%

Full history

30 of 2,000 profiles

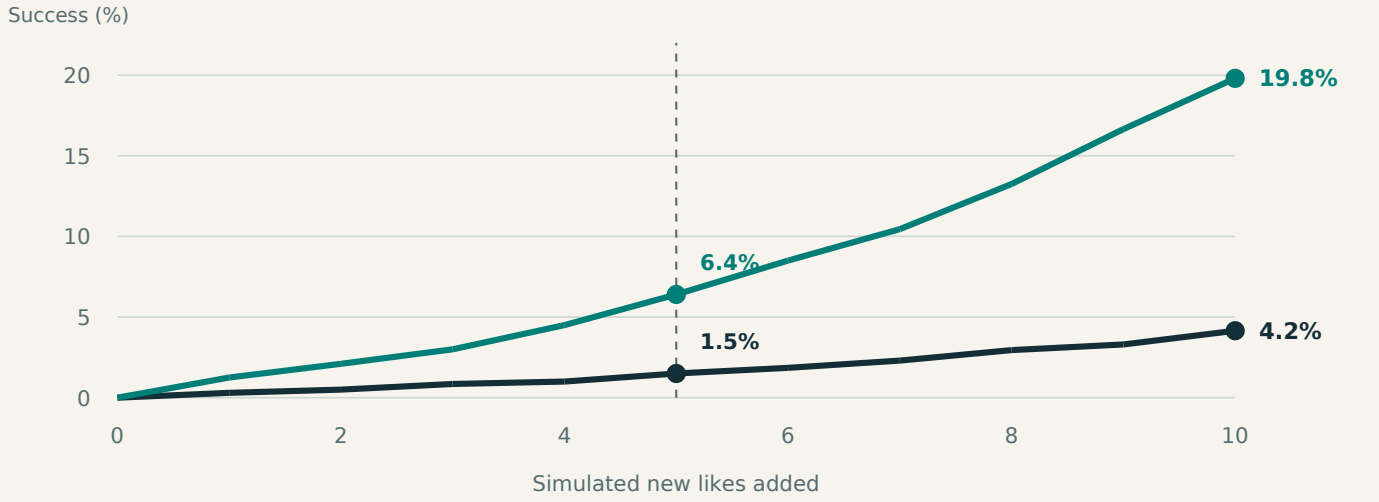
6.4%

Recent-20 reset

128 of 2,000 profiles

+4.90 percentage points after a reset. Paired 95% interval: 3.85 to 6.05 points.

MORE NEW LIKES HELPED, BUT MOST TRIALS STILL FELL SHORT



The test: 2,000 selected profiles received the same ten assigned target-genre likes under both methods. A reset keeps only the latest 20 starting likes, then adds the new ones.

Success: reach at least 3 target-genre movies in the top 10 at least once by that step. Both methods began below 3 by design. This measures a response to new signals, not proof of harm.

02 / WHAT GETS OFFERED?

Past interests shape the choices in front of you.

Before adding any new likes, we grouped steering-test profiles by their dominant historical genre. The full-history model offered each group a different genre mix.

1,955 of 2,000 profiles shown / Groups with at least 30 profiles

	Drama	Comedy	Action	Thriller	Crime	Adventure	Romance	Sci-Fi	Fantasy	Mystery	Other
Drama n = 1,301	31	10	8	10	9	5	7	5	3	4	9
Comedy n = 317	17	21	10	7	6	9	7	5	5	2	9
Action n = 168	10	5	21	12	7	15	2	12	5	3	6
Adventure n = 77	8	6	18	7	5	20	2	12	10	3	11
Thriller n = 51	21	7	15	15	6	8	6	6	2	4	10
Sci-Fi n = 41	10	6	20	9	6	15	1	17	7	3	7

Mean genre share of the top 10 (%)

0 35

Each movie is split equally across its genre labels before averaging. Rows total 100% before rounding; Other pools the remaining genres and any unlabelled share.

A detail worth testing: Sci-Fi-led profiles received 19.8% Action versus 16.6% Sci-Fi. Different exposure can reflect preferences and catalogue supply; it does not establish unfairness by itself.

NEXT Does personalization trade broader genre exposure for better predictions?

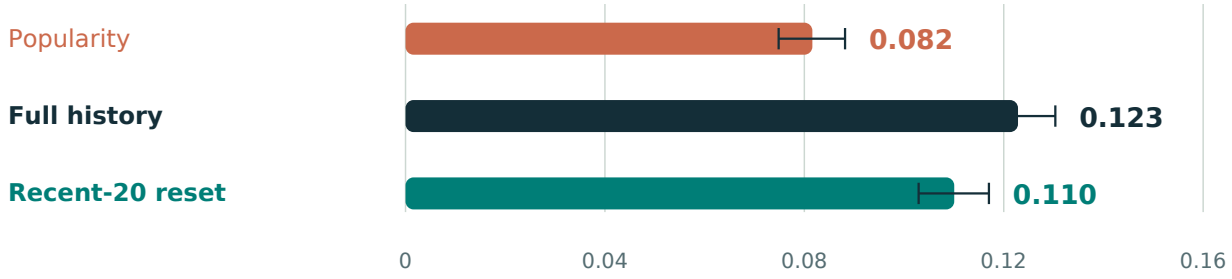
03 / THE TRADE-OFF

Easier to redirect. Not automatically more diverse.

A separate chronological test used real later likes, with no simulated signals. Full history predicted later likes best; the reset scored lower on both relevance and genre diversity.

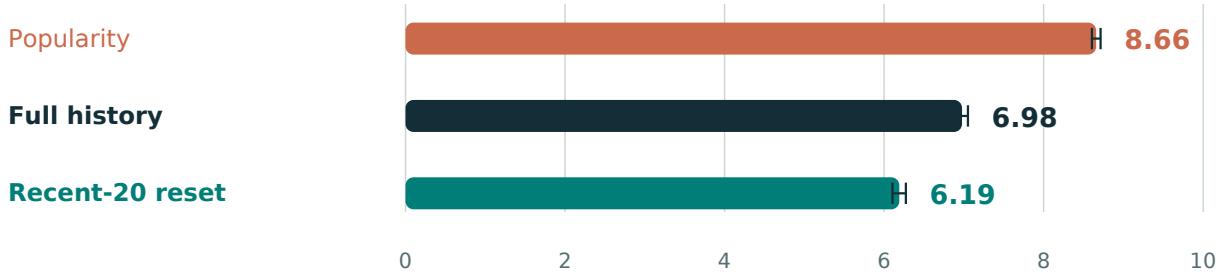
Relevance to later likes

nDCG@10, from 0 to 1. Higher means better retrieval of later likes.



Genre diversity

Effective genre count. Higher means a broader, more balanced genre mix.



Full history versus popularity: relevance rose by 0.041 (paired 95% interval 0.034 to 0.048), while effective genre count fell by 1.68 (interval 1.59 to 1.78).

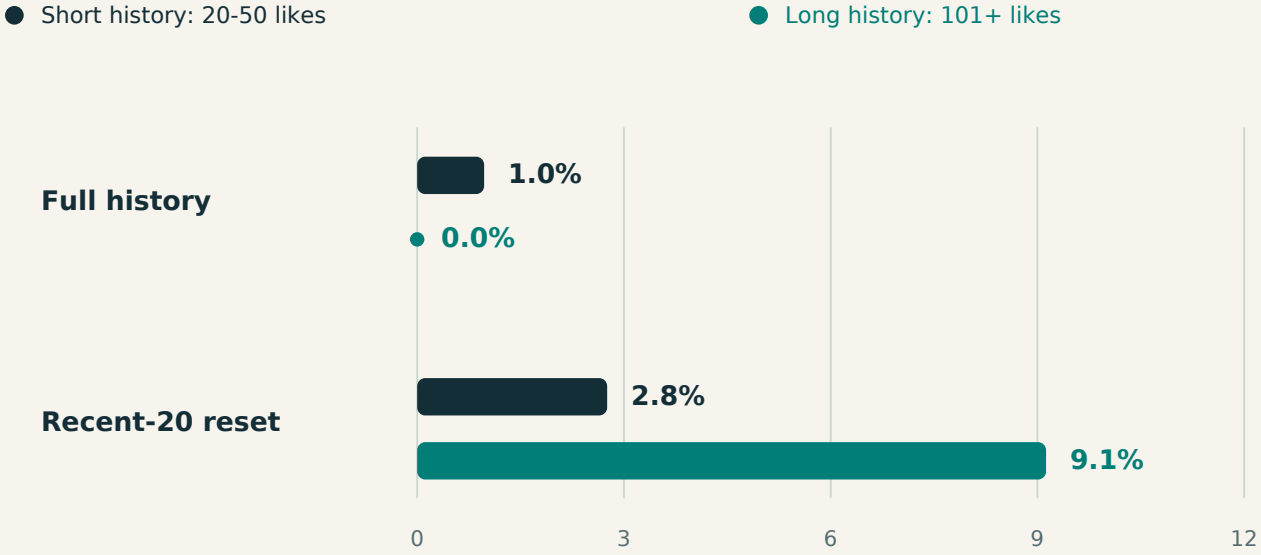
n = 2,000; Jan-Oct 2023 later likes. Bars are user means; whiskers are 95% user-bootstrap intervals. Genre labels measure content variety, not diversity of viewpoints. Unrated films are not known dislikes.

04 / WHO BENEFITS FROM A RESET?

The same reset. Different opportunities to steer.

We compared success within five new likes for profiles with short and long histories, using the same target-genre and starting-exposure weights for both groups.

STANDARDIZED SUCCESS RATE (%)



GAP: SHORT MINUS LONG

+1.0 pp
Full history

-6.4 pp
Recent-20 reset

Both groups became more responsive, but the absolute gap grew. A zero gap would mean equal rates on this measure; it could also mean equally poor responsiveness.

Gaps use unrounded rates; displayed percentages are rounded.

How we made tasks more comparable: 13 target-genre / initial full-history count strata, with at least 10 profiles per group in each. Retained: 685 short and 490 long histories; 825 profiles were outside this comparison.

Interpret cautiously: no full-history successes occurred among the 490 retained long histories. Sparse outcomes make naive bootstrap intervals unreliable; other differences remain unadjusted.

05 / WHAT SHOULD CHANGE?

Give users control. Show the consequences.

The design implication is a reversible profile control with a visible audit. Responsiveness, relevance, genre diversity and group differences should be evaluated together.

PROPOSED TOOL / A PROFILE CONTROL AND AUDIT PANEL

Let a user try full history or a recent-history reset, preview the resulting choices, and undo the change. Explain the retained history and show which outcomes improve or worsen.

DATA AND METHOD

Source. GroupLens MovieLens 32M: 32,000,204 ratings from 200,948 users. Only ratings, timestamps and movie genres were used. Source archive and CSV checksums were verified.

Training. Before 1 Jan 2023, retain movies with 100+ positive ratings (4+ stars). This gives 7,712 movies and 15,113,616 positive interactions from 197,301 users.

Model. 32-factor SVD on binary likes, with columns scaled by inverse square-root popularity. Normalize movie vectors; sum liked-movie vectors for each profile. Rank by dot product, excluding all rated movies. Movie vectors remain fixed.

Steering test. Select 2,000 profiles with 20+ prior likes and an eligible target: at most 10% of prior likes, 100+ catalogue movies, and fewer than 3 initial top-10 matches under both methods. Sample ten unrated target-genre movies uniformly; both methods receive the same sequence. Exclude injected movies from recommendations.

Exposure proxy. Generated top-10 recommendations, not observed views. Fractionally divide each movie across its genre labels; exclude IMAX and missing-label tokens. Effective genres = exp(-sum p log p), using known genres only.

Later-rating test. A separate random sample of 2,000 of 2,772 users with 20+ earlier and 5+ later positive ratings. Jan-Oct 2023 positives are test labels; none train the model. nDCG@10 rewards retrieving later likes near the top.

Uncertainty. 2,000 user-bootstrap resamples for steering and later-rating comparisons; paired differences use the same users. The model and one synthetic signal sequence per profile are fixed. Subgroup rates on page 4 are descriptive.

Limits and next validation. Selected volunteers, a popularity-filtered catalogue and selectively observed ratings limit generalization. Equal-weight sums dilute new signals in longer histories by construction. Resetting changes content as well as length. Compare random-20 histories, recency decay and repeated signal sequences before attributing the gain to recency.

No demographics, actual awareness changes or commercial-platform effects were measured. This is a test of one transparent recommender, not evidence that all feeds behave this way.

GroupLens / MovieLens 32M | Harper & Konstan (2015), doi:10.1145/2827872.
Related reachability research: Curmei, Dean & Recht (ICML 2021). Code, outputs and checks accompany this report.
Seeds: 20261018 (model/steering), 20261019 (extension). AI-assisted analysis and report preparation.